

Introduction to Bioinformatics

Microarray, NGS

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December 20, 2025

- Review of sequencing and reverse transcription
- Data cleaning, background correction, normalization
- Statistical theory: Negative Binomial, Limma, DESeq2
- Gene Ontology and biological pathways

What is RNA-Seq? (Expanded Definition)

- High-throughput sequencing of cDNA to measure transcript abundance
- Bulk RNA-Seq averages expression across many cells
- Captures gene expression, splicing, isoforms, and allele-specific expression
- Requires strong statistical modeling due to variability and bias

Sequencing Technologies – Detailed

- Illumina: sequencing-by-synthesis, low error, short reads (50–300bp)
- PacBio HiFi: long reads with high accuracy via circular consensus sequencing
- Oxford Nanopore: ultra-long reads (100kb+), real-time, higher raw error
- Choice affects mapping accuracy, isoform resolution, and cost

RNA-Seq Experimental Biases

- PCR amplification bias causing over-representation of some fragments
- GC-content bias affecting read distribution
- Batch effects from sample preparation and sequencing lanes
- Library size differences requiring normalization

Why data cleaning is necessary:

- Sequencing produces low-quality reads
- Adapter contamination and PCR bias
- Ambient RNA introduces background noise

Background Correction:

- Removes non-biological signals
- Corrects for random fragments and sequencing artifacts

Engineering Analogy: Removing baseline noise in a signal processing pipeline

Normalization: Making Samples Comparable

Problems with raw read counts:

- Different sequencing depths (20M vs 40M reads)
- Composition bias (highly expressed genes dominate)

Normalization Methods:

- Size-factor normalization (DESeq2)
- TMM normalization (edgeR)
- Voom mean-variance modeling (limma)

Goal: remove technical variation while keeping biological effects

Normalization – Deep Explanation

- **Goal:** make samples comparable despite sequencing depth/composition
- **DESeq2 median ratio method:** robust against highly expressed genes
- **TMM (Trimmed Mean of M values):** adjusts for composition bias
- **RLE:** uses median expression of stable genes

Why Statistics Are Essential in RNA-seq

- RNA-seq counts follow overdispersed distribution (NB $>$ Poisson)
- Mean–Variance dependence must be modeled
- Small sample sizes cause unstable variance estimates
- Statistical models maximize power while controlling false positives

Statistical Modeling: Why Not Gaussian?

RNA-seq counts are:

- Discrete (not continuous)
- Skewed
- Overdispersed ($\text{Var} > \text{Mean}$)

Negative Binomial Model:

$$K_{ij} \sim NB(\mu_{ij}, \alpha_i)$$

Variance:

$$\text{Var}(K_{ij}) = \mu_{ij} + \alpha_i \mu_{ij}^2$$

Why NB works:

- Captures extra variability
- More realistic for biological count data than Poisson

Negative Binomial Model – Foundation

- Variance = $\mu \times \mu^2 \times \text{dispersion}$
- Dispersion accounts for biological variability between replicates
- NB model fits gene-level count distributions
- Used in DESeq2 and edgeR

DESeq2: Hypothesis Testing Framework

Goal: detect differentially expressed genes

Null Hypothesis:

$$H_0 : \mu_{cond1} = \mu_{cond2}$$

Model: Negative Binomial GLM

$$\log(\mu_{ij}) = \beta_{0i} + \beta_{1i} \cdot condition_j$$

Key Components:

- Size-factor normalization
- Dispersion estimation and shrinkage
- Wald test or Likelihood Ratio Test

DESeq2 Statistical Pipeline

- 1 Size factor normalization.
- 2 Dispersion estimation and shrinkage via MAP
- 3 GLM fitting using NB distribution
- 4 Wald test or Likelihood Ratio Test (LRT)
- 5 FDR correction via Benjamini-Hochberg

Voom Transformation:

- Converts counts to log-CPM
- Estimates mean-variance relationship
- Produces precision weights

Weighted Linear Model:

$$y_{ij} = X_j \beta_i + \epsilon_{ij}$$

Empirical Bayes Shrinkage:

- Stabilizes variance estimates
- Similar to regularization in ML

Null Hypothesis:

$$H_0 : \beta_{1i} = 0$$

Limma + Voom: Statistical Pipeline

- 1 Convert counts $\rightarrow$ log2-CPM
- 2 Estimate mean-variance curve
- 3 Compute observation weights
- 4 Fit weighted linear model for each gene
- 5 Empirical Bayes shrinkage $\rightarrow$ moderated t-tests

Zero Inflation and Low-Count Genes

- Zero inflation occurs due to stochastic dropout or low expression
- Impacts false positive and false negative rates
- Models like ZINB-WaVE attempt zero-inflated NB modeling
- DESeq2 + shrinkage help stabilize low-count gene estimates

What Are Ontologies in Biology?

- Ontology = structured framework defining concepts and relationships
- Biological ontologies use controlled vocabularies for annotation
- Purpose: unify biological knowledge, reduce ambiguity, enable computation
- Example: Gene Ontology (GO) — hierarchical DAG structure

Gene Ontology (GO) – In Depth

- Three Categories: Biological Process, Molecular Function, Cellular Component
- GO terms describe roles, activities, and locations of gene products
- Terms connected by "is-a" and "part-of" relationships
- Used heavily for enrichment analysis after DE

GO Categories:

- **Biological Process** (e.g., cell cycle, apoptosis)
- **Molecular Function** (enzyme, receptor)
- **Cellular Component** (nucleus, membrane)

Analogy: A hierarchical “type system” describing gene roles.

Why GO Matters:

- Helps interpret long gene lists
- Clusters genes by function

Functional Enrichment Importance

- DE alone does not reveal biology
- GO and pathway enrichment connect DE genes to biological functions
- Helps interpret biological mechanisms, signaling cascades, pathways
- Used to generate hypotheses for wet-lab validation

What Is a Biological Pathway?

- Pathway = ordered set of molecular interactions producing a biological effect
- Represents metabolic, signaling, or regulatory flows
- Defined in databases like KEGG, Reactome, WikiPathways
- Pathways reveal the biological mechanisms influenced by DE genes

What is a Pathway?

- A biological circuit of interacting proteins/genes

Examples:

- MAPK signaling pathway
- Glycolysis
- p53 apoptosis pathway

Why Analyze Pathways?

- Reveals higher-level biological changes
- Easier than interpreting hundreds of genes individually

Engineering Analogy: Modules in a software system activated under certain inputs.

Types of Pathway Analysis

- Over-Representation Analysis (ORA): tests enrichment from DE gene list
- Functional Class Scoring: uses aggregated gene statistics
- GSEA: ranks all genes and evaluates directionality and distribution
- Topology-Based Analysis: considers network structure (e.g., SPIA)

Summary

- Sequencing converts RNA to cDNA and digital reads
- Cleaning + normalization remove technical noise
- DESeq2 uses Negative Binomial GLM + shrinkage
- Limma + Voom uses log-CPM + weighted linear models
- GO Pathways give biological meaning to gene lists

Best Practices for Teaching RNA-seq Concepts

- Illustrate count distributions and mean-variance dependence
- Show step-by-step workflow: FASTQ → DE → enrichment
- Use figures demonstrating normalization and dispersion effects
- Explain ontology hierarchy and pathway logic visually

References

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The End

Questions? Comments?